1. Describe the most relevant learnings from programming-related subjects during your career.

I am going to tell you about the programming-related subjects I took and the knowledge I gained from each of them.

<u>Algoritmos y estructuras de datos (Algorithms and Data Structures)</u>: This subject was taught using the Python language. It was the introductory programming course at university. It focused on learning the basic logic of programming: variable assignment, data types, different types of operators, control structures, and defining functions. We used the structured paradigm. We were evaluated with different practical exams, theoretical quizzes, and an integrating group project, which in my opinion was the most attractive challenge of the subject.

<u>Paradigmas de programación (Programming Paradigms):</u> In this subject, I learned that there are different programming paradigms. During the course, we saw the functional, object-oriented, and logical paradigms. We focused mainly on the object oriented paradigm with Smalltalk. In this subject, along with another one I was taking called Systems Analysis, I learned many concepts of the object oriented paradigm. I became able to abstract the entities of a domain. What I learned from the other paradigms, which we saw with Haskell and Prolog, is that in programming there are different ways to communicate with the computer and that each one has advantages or disadvantages depending on the problem we want to solve.

<u>Desarrollo de Software (Software Development):</u> In this subject, I learned the basics of web development. I learned about the division between the backend and frontend layers, and the communication between them through REST APIs. In the frontend, I learned HTML, CSS, and JavaScript, and then we also used React. For the backend, we also used JavaScript with the Node.js execution environment. I learned about creating APIs with Express.js, testing endpoints with Postman, consuming APIs with fetch and axios, and using ORMs. I learned that there are different ways to connect a database and the advantages and disadvantages of each in development and production. In this subject, we used a database in SQLite files. I also learned how to use npm.

<u>Backend de Aplicaciones (Backend Applications):</u> This is an elective subject. I feel that in this one I learned a lot through trial and error. We used Java. In this subject, I was able to apply what I learned about the Object-Oriented Paradigm. We learned file management, how to process a CSV (in fact, that was what the exam was about), and exception handling. What I liked most was the group project. We used Spring and Maven. We had to create a logistics control system. We used a microservices architecture with an API Gateway. We designed the database tables. We made an integration with the Google Maps API. The biggest challenge was making the different microservices communicate with each other correctly. In the end, we orchestrated the creation of containers with Docker. For all the functionalities, we prepared tests in Postman.

I would also like to talk about other subjects that were not directly related to programming, but from which I obtained useful knowledge.

<u>Análisis de Sistemas y Diseño de Sistemas (Systems Analysis and Systems Design)</u>: In these subjects, I learned everything about the Object-Oriented Paradigm. Also about the different types of software development processes, the 4 Ps in software development, and a lot about UML. Analysis and design patterns, SOLID principles, requirements engineering, and software architecture.

Lógica y Estructuras Discretas (Logic and Discrete Structures): Here I learned a lot about Boolean algebra, and first and second-order logic. These are things that I feel helped me a lot when learning to program.

Sistemas Operativos (Operating Systems): Besides learning about the basic structure of an OS, file systems, memory management, and processors, in this subject I learned how to use the command line.

Arquitectura de Computadoras y Sintaxis y Semántica de los Lenguajes (Computer Architecture and Syntax and Semantics of Languages): In these subjects, I learned about how computing works at a more basic level, closer to the hardware.

Bases de Datos (Databases): I learned mainly about relational databases, SQL queries, normalization, DB indexing, and the parts of a DB.

2. Describe a situation you had to resolve with a teammate from a study or work group.

During the development of the project for my System Analyst thesis, I faced some situations. The project consisted of an order management system for a factory for a pharmacy/laboratory. Two other teammates and I were in charge of designing the database. One teammate was determined to propose a solution that was clearly not correct, as it violated the principles of normalization. He got angry. To solve this, I decided one day to talk privately with him. We reviewed what we learned in the Database subject about Normal Forms and fortunately, he was able to understand the disadvantages of what he was proposing. We managed to reach a solution, with all team members satisfied and without anyone feeling undervalued (this is something we talk about in our weekly meetings).

3. Describe your personal learning plans or development expectations in technical areas in the short and long term.

Short term:

- Continue with my gym application project. I want to keep working on my personal project. I recently finished the module that uses AI to generate general routines, which can be accessed by scanning a QR code on a gym wall.
- Learn in a real-world environment. I would like to learn in an organized and well-structured company. I have high expectations about getting this internship to understand how large-scale projects work and to learn from more experienced developers.
- I will also work hard to learn as much as I can at university and make the most of this academic year.

- Currently, I am teaching myself Node.js and Django through online courses and other resources available on the web.

Medium/Long term:

- Graduate as an Engineer. My main goal is to get my degree in Information Systems Engineering.
- I am interested in how applications scale and run efficiently. Therefore, I’d like to learn DevOps, Cloud Computing, and Server management. I want to learn how to manage cloud environments and understand the full lifecycle of software, from development to deployment.